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Hybrid Modeling for Bioprocesses: Architectures, Applications, and Perspectives

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ABSTRACT

Recent hardware developments within industrial digitalization make more evident the need of high-fidelity models to exploit process system engineering potential in biomanufacturing. Hybrid models, which combine first-principles (white-box) and data-driven (black-box) approaches, aim to leverage the strengths of both methodologies to mitigate their individual limitations to provide high predictive power. This article explores the integration of hybrid modeling techniques in bioprocess engineering, emphasizing their potential to facilitate product development, accelerate experimental design, enhance process monitoring, drive optimization and catalyze the implementation of digitalization strategies to achieve a more robust and sustainable production in the framework of the goals set by the UN 2030 agenda. In this review, we analyze 270 publications from the early 1990s to 2024, extracting application domains, hybrid-model architectures, implementation software, and nonparametric output structures. Our findings highlight marked improvements in predictive accuracy, and extrapolation capabilities while revealing persistent challenges in integrating mechanistic and empirical components. This review on hybrid modeling implementations shows their potential to enhance accuracy, calibration, and extrapolation capabilities, highly required for bioprocesses. This work underscores the importance of integrating mechanistic and data-driven approaches to improve the flexibility and interpretability of models in complex engineering systems, offering significant implications for more sustainable and innovative bioprocesses.

1 | Introduction

Sustainability has become a critical focus in addressing global challenges, as it ensures that current development meets the needs of present generations without compromising the ability of future generations to meet their own needs. The Sustainable Development Goals (SDGs), set by the United Nations, provide a comprehensive framework to tackle pressing issues such as climate change and environmental degradation. Among these, goals like Industry, Innovation and Infrastructure (SDG9), Responsible Consumption and Production (SDG 12), Climate

Action (SDG 13), and Life Below Water (SDG 14) are directly impacted by how industries operate, making sustainable practices essential [1].

As of 2024, the SDGs were in a dire situation, with most of them being severely delayed and targets not on track to being met [1]. For SDG 9, less than 20% of the targets were on track to being met, whereas for SDG 12 the share was around 30%. Moreover, the UN estimates that 84% of the SDG targets have either stagnated or shown a reversal of progress. This indicates that further acceleration is needed to achieve the targets. Research in diverse

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areas of knowledge and articulation with industry is crucial to meet the 2030 agenda [1].

In the framework of the above-mentioned goals, biorefineries emerge as an attractive approach to achieve them. Second and third generation biorefineries (2G and 3G) allow biomass and agricultural wastes to take full potential, without compromising food security, which in turn makes them very attractive for adding value to waste [2]. Bioprocesses offer a promising avenue to meet sustainability goals by reducing reliance on fossil fuels and lowering carbon footprints [3]. However, they also pose significant challenges, including the complexity of biological systems and the need to balance economic viability with ecological responsibility [4]. Moreover, biorefineries are often complex processes that involve many steps, posing a technical challenge from their inception through the implementation.

One of the main challenges of biorefineries takes place in the bioreaction stage. Biotransformations are intricate, and their mechanisms are usually not well understood, which leads to low titer, productivity, and yield.

Process systems engineering (PSE) plays a vital role in addressing these challenges by integrating knowledge from engineering, mathematics, and computer science to optimize bioprocesses [5]. PSE methods enable the design, control, and operation of bioprocess systems in a way that improves efficiency and minimizes environmental impact. For instance, through advanced modeling, simulation, and optimization techniques, PSE can help predict the behavior of biological systems, optimize resource usage, and identify bottlenecks in production [6, 7].

In PSE, modeling takes a central role for process simulation and its use [8, 9]. Models can administrate information from two main perspectives: structural (our understanding of the underlying mechanisms) and the model parametrization (the nature of the numeric techniques used to describe the relationships between data). From the structural point of view, first-principles, mechanistic or phenomenological models represent a wide class of models, usually called white-box, due to being developed primarily on conservation equations and physical laws. First principles models are constructed through deductive reasoning, wherein the model equations are derived from existing knowledge of the system, using established relationships such as conservation laws. Mechanistic models usually comprise a system of differential algebraic equations (DAEs). An important characteristic of mechanistic models is that they can be extrapolated to process conditions outside of the calibration space, thanks in part to being developed from conservation laws, although this is not always the case [10]. In contrast, data-driven models typically employ an inductive approach, wherein a generic mathematical model is fitted to map inputs to outputs, without considering the intrinsic structure of the phenomena, unlike their white-box counterparts [10–12]. Data-driven techniques usually take the form of arbitrary functions to model the relations between inputs and outputs. Gray models usually fall in between, combining mechanistic knowledge and data-driven techniques [13, 14].

From another perspective, models can be classified regarding the form of model parametrization. Parametric models are established beforehand based on the knowledge of the process [15].

These models possess a finite number of parameters, often characterized by physical significance, which vary according to the model's complexity [11]. In contrast, nonparametric models are solely derived from data, lacking predefined parameters [15]. The term “nonparametric” does not mean that the model is built without parameters, but that the number of them is variable, not fixed in advance by knowledge, and that they commonly lack physical significance [11]. In nonparametric modeling, mathematical techniques such as algebraic, series expansions, machine learning (ML), and other black-box approaches are used to derive insights from inputs to outputs. These techniques are also called “surrogate models” when they are used to approximate the behavior of more complex and computationally costly models [16] and are used to describe the relevant process behavior exclusively from model-generated data, to perform calculations faster such as optimization [17, 18]. Typically, they have great predicting power inside the space of their calibration, but extrapolation is not advisable [19].

For a better understanding of the advantages and disadvantages of both parametric and nonparametric modeling, Table 1 shows a comparison between parametric and nonparametric modeling [20–27].

Table 1, presents advantages and limitations between parametric and nonparametric modeling [20–27]. Parametric models are built based on prior knowledge of the system, incorporating physical laws and constraints, which makes them highly interpretable and ensures their compliance with conservation laws. They have low variability, making them more robust and reliable, especially when data is scarce. Parametric models exhibit strong extrapolation capabilities, allowing them to make reasonable predictions beyond the range of the observed data. However, their reliance on predefined equations introduces a strong bias, which may lead to inaccurate predictions if the assumed structure does

TABLE 1 | Comparison between parametric and nonparametric modeling [20–27].

	Parametric modeling	Nonparametric modeling
Model variability	Low	High
Model bias	High	Low
Model parameter data required	High	Low
Mechanism understanding required	High	Low
Model flexibility	Medium	High
Explainability and interpretability	High	Low
Interpolation capabilities	Medium	High
Extrapolation capacities	Good	Bad
Computational training cost	Low	High
Risk of overfitting	Low	High
Compliance with conservation laws	High	Low

not fully capture the system dynamics. Developing parametric models requires extensive domain expertise, making the process time-intensive and complex, often involving experts from multiple disciplines.

Nonparametric models offer greater flexibility as they do not require prior mechanistic knowledge, making them particularly useful when the underlying system is poorly understood or difficult to describe mathematically. These models can efficiently capture complex, nonlinear relationships within data, often uncovering hidden patterns that parametric models might miss. However, their flexibility comes at a cost. Nonparametric models typically require large amounts of high-quality data for training, and their computational cost is significantly higher, particularly during the training phase. Additionally, they are prone to overfitting and overparameterization, making them less robust, and they often struggle with extrapolation beyond the observed data range. They are usually not reliable enough to extrapolate to conditions outside of their calibration space; however, they have great interpolation capabilities within it. Another critical drawback is their lack of interpretability, as they do not rely on physical constraints or conservation laws, which can make it difficult to understand the underlying mechanisms driving predictions.

From previous experience, there is no ideal model, with both parametric and nonparametric models having strengths and shortcomings that not only depend on system understanding, but also on data and computational power. As a potential trade-off, the scientific community has proposed hybrid modeling, also called hybrid semi-parametric modeling, to overcome the challenges that arise from parametric and nonparametric modeling, while retaining most of their advantages.

Notice that researchers do not always agree on the definition and use of the term “hybrid model.” Some authors claim to have developed hybrid models, when they combine different mechanistic and empiric knowledge (mathematical relationships that provide reliable approximations but are not built using mechanistic knowledge) such as correlations for heat and mass transfer coefficients, for example, [28–34]. Besides, the term has been used when modified structures apply to pure nonparametric approaches for predictions. These modifications usually entail combinations of artificial neural networks (ANNs) with other ML techniques or optimization algorithms [35–40]. Finally, “hybrid modeling” has been applied to other approaches such as Computational Fluid Dynamics applications that integrate various numerical methods [41] and process system engineering simulations that use multiple software tools to perform the simulations [42–44]. Here in, “hybrid modeling” represents a modeling paradigm that seeks to combine the strengths of both parametric and nonparametric modeling while minimizing their individual disadvantages.

Hybrid modeling is an emerging approach gaining traction; thus there are recent reviews in the literature detailing the applications of hybrid modeling to separation processes [45], biochemical processes using neural networks [24, 46, 47], wastewater treatment (WTT) [48]; chemical, petroleum and energy systems [46, 49]. For bioprocesses specifically, recent reviews include topics such as the role of hybrid models in bioprocess scale-up [50]; the applications of hybrid modeling to systems biology for mammalian

cell culture development [51], and the prospects of hybrid modeling for industry 4.0 and 5.0 [20, 23]. Those contributions delve into the details of relevant discussions. However, they lack quantitative and qualitative trends and information on how hybrid models have been implemented in bioprocesses. This study provides the main characteristics of hybrid models, such as structure, knowledge integration, performance assessment, data requirements, and challenges. This contribution focuses on the hybrid modeling of reactions in bioprocesses, aggregating information from 270 contributions screened down from 5890 identified from a bibliometric search. From there, the target is to identify trends such as the types of bioprocesses to which hybrid modeling has been applied, the preferred architectures/structures/design patterns, the most used nonparametric techniques, the scale of the experimental data used to develop the models, the mode of operation of the experiments, and software used for building the models, as well as cross-correlations between these categories.

This study describes the literature review methodology in Section 2, Section 3 presents the general characteristics of hybrid semi-parametric models, whereas Section 4 shows insights and correlations of the different implementations of hybrid models to bioprocesses, including an overview of some recent applications of hybrid semi-parametric models to bioprocesses. Finally, Section 5 presents conclusions, current challenges, and future perspectives.

2 | Literature Review Methodology

A comprehensive bibliometric study was performed to quantitatively and qualitatively map the research landscape using descriptors such as journals, publication outputs, institutions, authors, and keywords. Full details of the methodology for the literature review can be found elsewhere [52]. Here, a summary is presented.

The search strategy began with identifying the need for information, focusing on hybrid modeling applications in bioprocesses. Non-biotechnological processes were excluded to maintain the relevance of the field. Scopus and Web of Science were selected as the primary databases due to their extensive and complementary coverage of scientific literature. A search equation was designed by expanding terms, incorporating keywords related to hybrid models (e.g., “hybrid model,” “hybrid neural network”) and bioprocesses (e.g., “bio*,” “bioproc*”). The search was restricted to publications from 1992, marking a seminal work in the field, through March 2024. The search was repeated on December 2024 to encompass new publications.

Search results were refined by targeting specific fields: “Title-Abstract-Keywords” for Scopus and “Topics” for Web of Science. Abstracts were validated manually to ensure relevance to Chemical and Biochemical Engineering, excluding unrelated documents. This iterative process included refining the search equation multiple times, guided by insights gained during analysis. The search continued until results stabilized, with less than 5% variation in top authors, journals, and total results across iterations. A total of 5890 contributions were found, and after a manual review, 270 contributions were selected as of

interest. Information was refined using software like Vantage-Point, VOSViewer, Bibliometrix, and Python. Later, a manual assessment of each contribution was performed.

3 | Hybrid Modeling Framework

This section is divided into subsections to provide a clear and comprehensive look at hybrid modeling. Section 3.1 introduces the general characteristics of hybrid models, laying the groundwork for understanding their key features. Section 3.2 dives into the different structures commonly used to implement these models, whereas Section 3.3 explores various ways to incorporate knowledge into their design. Section 3.4 highlights some of the most popular nonparametric substructures used in hybrid modeling for bioprocesses. Section 3.5 focuses on how the performance of these models has been assessed, followed by Section 3.6, which discusses the data requirements necessary for building them, and lastly, Section 3.7 showcases some of the main applications of hybrid models.

3.1 | Main Characteristics of Hybrid Modeling

In the middle of the parametric and nonparametric modeling paradigms, the called “hybrid models” or “hybrid semi-parametric models” exist, combining both first principles and data-driven techniques to describe the systems. Figure 1 shows the knowledge basis and mathematical representations for parametric, nonparametric, and hybrid modeling. Parametric models heavily rely on first principles and prior knowledge, using conservation laws, mass-energy-momentum balances, kinetic mechanisms, thermodynamic relationships, and transport laws

to describe the system. Parametric models are usually expressed as algebraic equations, DAEs or partial differential equations (PDEs), with data being mostly used for parameter identification (hence the dotted green line in Figure 1). Nonparametric modeling approaches are usually data-hungry [23, 26, 45, 50] and build their knowledge exclusively from process data, extracting no information from first principles. Nonparametric models are usually expressed as data-driven techniques such as ANNs, Gaussian Processes (GPs), and support vector machines (SVMs) or Autoregressive models (ARX, ARMAX, ARIMA). Hybrid modeling aims to compromise both approaches in the middle, needing fewer data than nonparametric models, but without requiring prior knowledge as extensive as parametric models do. Hybrid models can be expressed in a wide range of structures, mixing the data-driven techniques used for nonparametric modeling and the mechanistic knowledge expressed in parametric models. It is important to note that hybrid models do not necessarily use their parametric and nonparametric components in the same ratio. Thus, the spectrum seen in Figure 1 signals that hybrid models can be “darker” if they rely more on the nonparametric techniques than the mechanistic ones, or “lighter” if the inverse is true.

Literature often claims that hybrid modeling leverages most of the advantages of parametric and nonparametric modeling while minimizing its disadvantages. Although this is empirically seen in an increased performance when compared against full mechanistic models, there are sparse mathematical proofs. It has been proven that giving more structure to the nonparametric subcomponent via an additive term helps to reduce the asymptotic variance of the parametric parameters [53]. Authors prove that if the parametric model is ill conditioned, hybrid models will outperform it. Furthermore, a Focused Information Criterion is

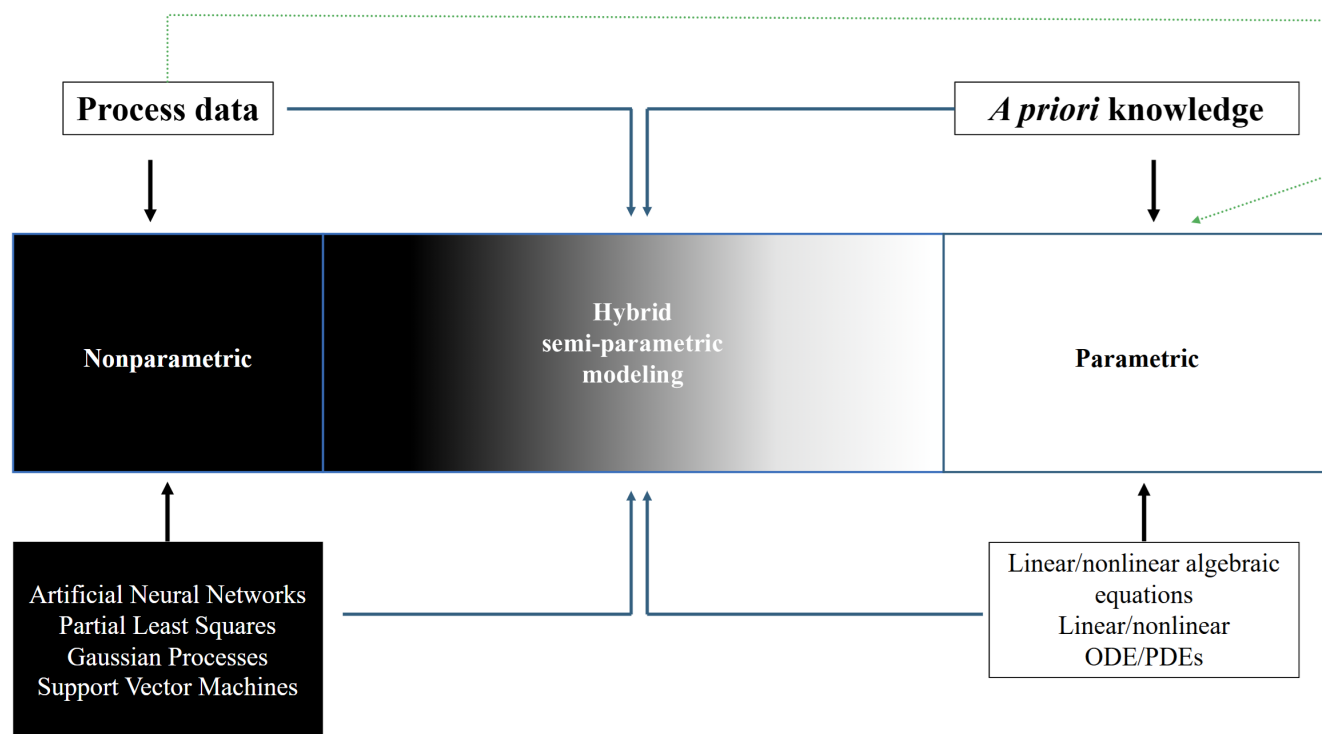


FIGURE 1 | Parametric, nonparametric, and hybrid semi-parametric modeling and the types of knowledge they are based on. Adapted from [11].

proposed, showing that in small data samples hybrid models are heavily reliant on induced bias from parametric ones, but if the data has low variance, the hybrid model enhances prediction capabilities requiring less data than a full nonparametric approach [54].

Hybrid modeling main characteristics are [25, 26, 49, 55]:

1. Hybrid models usually present better parameter estimation, due to:
 - a. Less parameters than pure nonparametric techniques, which can lead to higher statistical confidence of the identified parameters.
 - b. The chance to map all measurable quantities as inputs to the model structure, where the relationships are established using the nonparametric sub model. Thus, more information is included than in parametric models, usually allowed due to deficiencies in process understanding.
 - c. Combining data-driven approaches with mechanistic knowledge results in a constrained space that imposes physical meaning and bounds to the variables, which is hard to achieve in pure non-parametrical techniques.
 - d. When compared to pure nonparametric models, hybrid modeling exhibits superior calibration properties and enhanced extrapolation properties when compared to pure nonparametric techniques due to the structuring of the operation space using mechanistic or phenomenological knowledge. This structuring reduces the search space, consequently requiring less data for training hybrid models than for pure nonparametric approaches. However, the strategies for training and validation of hybrid models are heavily dependent on the model structure/architecture chosen.
2. Less mechanistic or first principles knowledge is needed for developing a hybrid model than a complex pure parametric model.
3. Hybrid models are usually developed faster than pure parametric models, since less mechanistic knowledge is

required. However, they are highly dependent on quality data and its availability.

4. The mechanistic knowledge enhances the model interpretability, making it more “transparent” than nonparametric models (hence the term “gray box model” is also used).

3.2 | Hybrid Model Design Patterns

Recently, a comprehensive overview of design patterns for hybrid models was presented [26]. These design patterns generalize, unify, and formalize diverse modeling approaches across various domains and applications. The following sections introduce these patterns and illustrate them through selected relevant studies from the literature. Notice that the chemical engineering literature has used the term “architecture” [11, 23, 56]. Recently, a distinction between “design pattern” and “architecture” was proposed because any model architecture can have different combinations of different design patterns, working in parallel, in series or nested inside another [26]. In the present work, “architecture” and “design pattern” are synonyms, whereas “structure” is used to discuss the different combinations and arrangements of the design patterns. It is important to note that a model structure can comprise one or several design patterns.

Agarwal [56] presented the general architectures in which the white box and black box components of a hybrid model can be arranged. Figure 2 shows three configurations. Configurations (a) and (b) show a serial architecture, where a black box output serves as input for the white box, and the white box output serves as input for the black box, respectively. Configuration (c) shows a parallel architecture, and finally, configuration (d) shows a data-driven technique constrained by a parametric model. Configurations (a), (b), (c), and (d) are also called “feature learning,” “physics-based preprocessing,” “delta model,” and “physical constraints,” respectively [26]. In Figure 2 there are two types of inputs for configurations (a), (b), and (c), named x_1 and x_2 (color coded blue and green, respectively) to clarify that the inputs for the different subcomponents of hybrid models might be different, but also could include common variables.

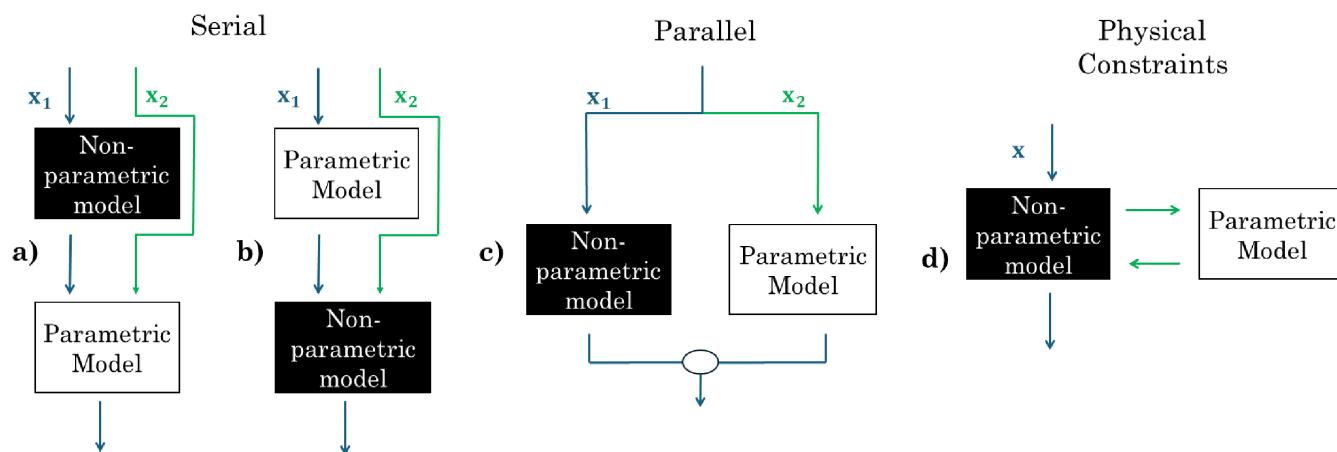


FIGURE 2 | Types of design patterns used for building hybrid models: (a) feature learning; (b) physics-based preprocessing; (c) delta model; (d) physical constraints Adapted from [11, 26].

The design pattern choice not only influences the interaction between the different components of a hybrid model, but also determines the training strategy and, therefore, computational cost of developing a model.

3.2.1 | Serial Architectures

The most popular hybrid model architecture is the “feature learning” design pattern. In this architecture, the white box is usually composed of conservation laws, mass, heat, or momentum transfer relations that describe the process. The black box typically represents the kinetics of the process or the transport coefficients, for which there is no general valid model [10, 47, 57]. This architecture is very useful when there is no precise knowledge of the underlying mechanisms, but there is enough data to identify the underlying patterns. Another scenario is presented when there is a huge amount of data that can be exploited to model the underlying mechanisms [58, 59].

The “physics-based preprocessing” design pattern is mostly used in data sparse environments, primarily using data augmentation [12]. In this architecture, the white-box model predictions are used to establish a link between the state of the system and certain parameters [11]. Some examples include: the prediction of cell culture using ANNs [60] where authors used an ANN to model some reaction rates in their dynamic balance flux model using other mechanistic reaction rates and states and inputs to the data-driven components, a fed-batch process for producing penicillin [61] taking advantage of Kalman filters to estimate and reconstruct some states from noisy data and predict fault detections, the control of biomass in protein production using *Pichia pastoris* [62] implementing a soft-sensor to predict biomass concentration based on online measurements of CO₂ utilizing a multi-linear regression that takes advantage of the mechanistic carbon balance.

When utilizing serial architectures, the training can be done in two different ways: coupling the identification of the parametric and nonparametric components simultaneously or decoupling the identification, usually calibrating a parametric structure and then calibrating a nonparametric component to account for the deficiencies of the parametric component.

3.2.2 | Parallel Architecture

The parallel architecture (“delta model” design pattern) applies if there is a process model (white box) that captures the underlying mechanisms but has limited predictive power due to structural factors not accounted for (unmodeled effects, nonlinearities, dynamic behavior) [26]. The predictive power of the model can be enhanced using nonparametric techniques, although their performance remains poor outside of the training region [11, 23]. This structure is interesting if the system can be decoupled into non-interacting parts, that is, static non-linear and dynamic linear behavior. This modeling approach has been applied to cases such as predicting phosphorus removal in a WTT process [63] correcting the model’s mismatch by adding the output of an ANN to the mechanistic reaction rates to enhance prediction capabilities; using Kalman filters to correct bias from mechanistic model

predictions and enabling higher forecasting capabilities in each integration time in ethanol fermentations for biofuel production [64]; using ANNs to correct the predictions of parametric models in cell apoptosis [65]; applying corrections to the activated sludge model in an industrial cokes wastewater-plant [66] correcting the activated sludge model by using five different data-driven techniques like ANNs and multiple linear regressions; predicting biomass concentration in a baker’s yeast fermentation [67] utilizing a mixture of experts, where each data-driven component is used to predict the states and a gating function combines them for enhancing prediction capabilities; studying polymerization processes by simplifying the parametric structure of the model [68] or fed-batch penicillin fermentations [15] where simplified models are proposed and the data-driven techniques take full advantage of residuals and state predictions to have similar performance to more computationally demanding models.

Researchers have proposed three main ways for approaching the parallel structure, shown in Figure 3 [11]. The (a) configuration is called “superposition” where the black box output is summed or subtracted from the white box model prediction to enhance the model performance. This configuration works as a bias correction, changing the output variance and shape. This is the most used parallel structure, first proposed in bioprocesses to study a fed-batch penicillin fermentation [15]. Configuration (b) is called “multiplication” because the output of the model is the multiplication of the outputs predicted by the black box and white box, working as a correction factor (amplification or mitigation of the parametric outputs). It was first utilized in bioprocesses for a bakers’ yeast production process [17]. The (c) configuration is applied when the outputs of the black box and white box are transformed by either a black box or a white box model [69], as such, weight factors (w_1 and w_2) are applied to combine them and obtain the final prediction. It is important to note that the inputs to the nonparametric and parametric parts could be different (as evidenced in Figure 3); therefore, the nonparametric component might obtain knowledge from information that the parametric models do not consider in their formulation.

3.2.3 | Physical Constraints

Physical constraints represent a hybrid modeling design pattern that integrates domain knowledge—such as conservation laws—into the structure, parameters, or outputs of a data-driven model [20, 26, 70]. These constraints can be categorized as hard constraints and soft constraints, based on how strictly they are enforced.

- Hard constraints encode physical principles directly into the model’s structure, ensuring they are never violated. For instance, adding a ReLU (rectified linear unit function, which is a function that is 0 when its input is less than or equal to 0 and is a linear function when its input is greater than 0) activation layer to guarantee non-negative outputs for variables like concentrations or pressure, or using a softmax layer to ensure molar fractions sum to one, are common examples [71]; or using Neural ODEs to predict ethanol production via fermentation [72] or Chinese Hamster Ovary cells (CHO) biomass growth in a bioreactor [73]. However, hard constraints require the input data to align with the

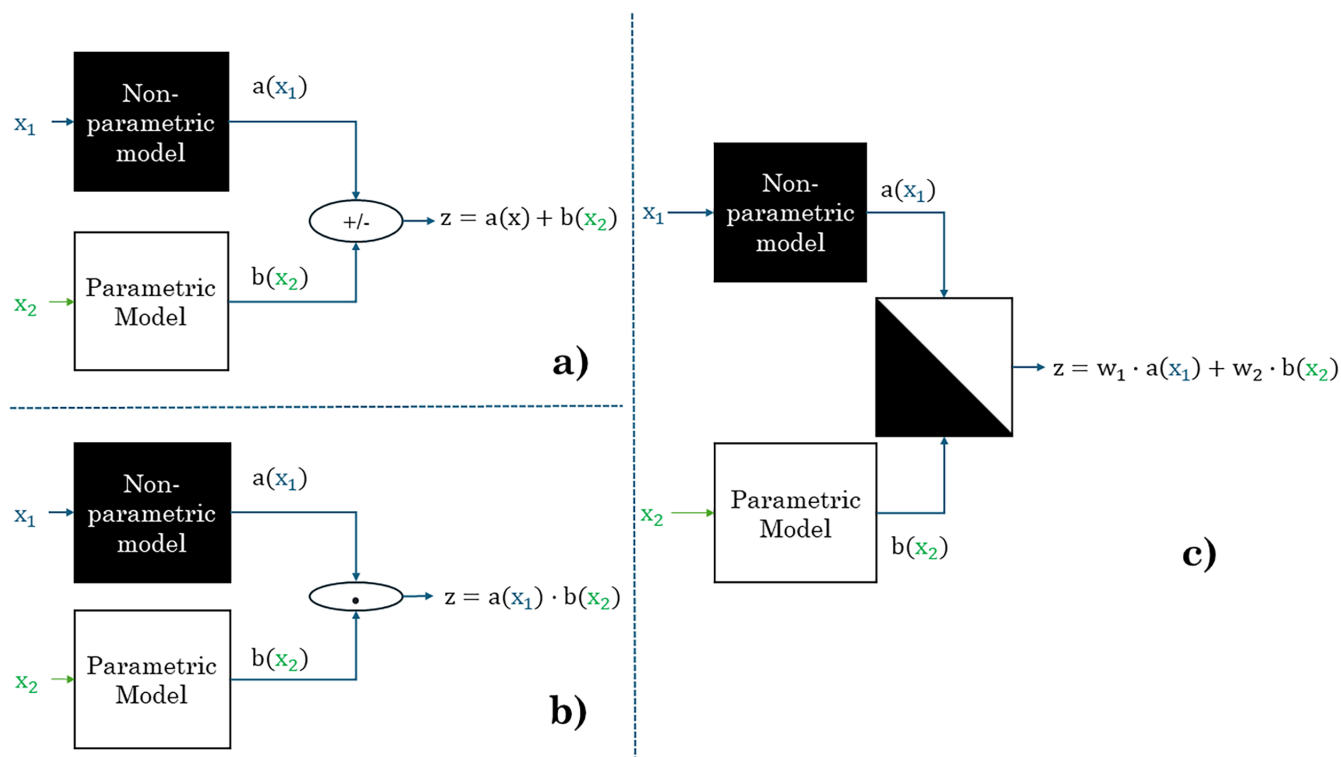


FIGURE 3 | Types of design pattern architecture. (a) linear superposition; (b) multiplication; (c) weighed predictions by using either a black box or a white box. The individual predictions of white boxes are denoted as $b(x_2)$ while for black boxes they are denoted as $a(x_1)$. In configuration (c), the individual predictions are added using weighting factors (w_1 and w_2). Adapted from [11].

imposed rules, a process known as data reconciliation [20]. A challenge with hard constraints is that they can result in non-differentiable models complicating gradient-based optimization methods [26].

- Soft constraints incorporate mechanistic insights into the loss function as penalties. These penalties guide the model during training to produce outputs that align with domain knowledge, allowing some flexibility in how strictly the constraints are followed [20, 26]. Once training is complete, the model relies solely on the data-driven architecture, with the domain knowledge encoded implicitly. Physics-informed neural networks (PINNs) are an example of soft-constrained models [70]. PINNs have been successfully applied to simulated cases of bioprocesses [74, 75]; kinetics of biodiesel production [76]; and WTT plants [75].

When using hard constraints at the output, the results are guaranteed to be physically plausible; however, these constraints can introduce non-differentiable nonlinearities that complicate gradient-based optimization. To address this, soft constraints are sometimes used because they provide a smoother optimization landscape, even though they do not guarantee exact adherence to the desired conditions. Moreover, combining multiple physical constraints in a single model can be challenging. A key benefit of incorporating physical constraints is enhanced data efficiency. By embedding prior knowledge into the model, the complexity of the problem can be reduced, which in turn lowers the amount of training data needed [26].

3.3 | Knowledge Integration

The standard way to incorporate mechanistic knowledge into a hybrid model is using mathematical expressions such as conservation laws or underlying phenomena. But they are not the only form of knowledge that can be incorporated. Some examples are:

- Knowledge about different stages of a process (e.g., growth phase; lag phase; etc. for a fermentation) can be leveraged to apply fuzzy logic (FL) either to enhance prediction for each phase by changing the mechanistic model [77–80] or to change the structure of ANNs [81].
- Design of experiments can evaluate and optimize the black box structure [82].
- Constraints can be added to the output predictions of the hybrid model to enhance its performance [25].
- Physics laws can be encoded in the training phase of the black-box component as in Physics Informed Neural Networks [47].

Mechanistic knowledge should be incorporated only if it is highly certain, which means that it does not contain empirical assumptions to avoid producing a model with high bias. The white box should be limited to knowledge that represents the physical interactions of the system [25]. For example, mass balances can be incorporated without hesitation, as the mass conservation laws apply to all systems. Assumptions like lumped parameters in a

chemostat might not always hold true, especially in large systems; therefore, the assumption introduces bias to the model.

The unknown elements of a hybrid model, such as kinetic terms, unknown or unmeasured variables, or mass transfer coefficients are usually represented by a nonparametric structure.

3.4 | Data Driven Components of Hybrid Models

Since the beginning of the century, the increasing availability of data, driven by the continuous advancement of better and more precise sensors, larger databases, and enhanced computational power, has compelled engineers to recognize data as a valuable source of information [12]. Therefore, mathematical techniques that build insights from data are highly valuable. Non-parametric and data-driven techniques can be broadly classified in the following categories, with accompanying examples of their applications to bioprocesses:

- Algebraic expressions such as polynomials or rational functions for predicting the pH and acetic acid production of Kombucha fermentations [83], and sigmoidal and exponential functions for describing methane production during the anaerobic digestion of orange peel waste [84].
- Series expressions such as Fourier series used for optimizing the pH profile for controlling the process of the yeast *Wickerhamomyces anomalus* against the spoilage yeast *Zygosaccharomyces rouxii* in wine pre-fermentation [85].
- Statistical models such as principal component analysis (PCA) for identifying the correlations between variables for monoclonal antibodies production [86], and multivariate data analysis (MVDA) for predicting the production of adenovirus using human embryonic kidney cells (HEK 293) [87].
- Time-series autoregressive approaches such as ARMAX for controlling dissolved oxygen in a *S. cerevisiae* fermentation [88], ARIMA for modeling and predicting a membrane bioreactor (MBR) for treating ampicillin from pharmaceutical wastewater [89] or ARX for predicting biomass concentrations in a lactic acid fermentation [90].
- ML approaches such as ANNs, as well as techniques like GPs or SVMs [91]. An overview of several nonparametric structures such as ANNs, Random Forests (RF), SVMs, GPs, among others has been recently presented [92].

The choice of nonparametric technique coupled with the hybrid model's structure heavily influences the development and the strategy for training and validation of the model.

3.5 | Hybrid Model Performance Assessment

Hybrid models' performance and extrapolation capabilities are directly related to the structure of the nonparametric portion of the model. This refers not only to whether the architecture is parallel or serial but also to the configuration and the specific model structure of the nonparametric model, particularly its size (e.g.,

layers of an ANN, number of neurons per layer, type of activation function, etc.) [11].

The size of the data-driven technique can be assessed using the Akaike Information Criteria (AIC) or Bayesian Information Criteria (BIC), the latter being preferred for models with a high number of parameters [55, 57]. For discriminating between models, Friedman's test can be applied [93]. In general, there should be a trade-off between estimation quality and the number of parameters, to avoid overfitting or underfitting issues, which should also be addressed in training and validation.

To compare hybrid models' performance to traditional parametric and nonparametric models, researchers have also used more conventional metrics such as the coefficient of determination r^2 despite not being suited for nonlinear estimators in biochemical systems [94], the mean squared error (MSE) and its derivatives such as root mean square error (RMSE) or normalized MSE (NMSE), error measurements such as mean absolute percentage error (MAPE), mean absolute error (MAE), among others.

When extrapolating the hybrid model's predictions outside the training domain, some assessment of the validity of the extrapolation should be done. Authors have mainly used two complementary criteria: convex-hull criteria and a confidence interval criterion. The convex-hull criteria are used to verify that each empirical model solely interpolates the data gathered during model identification. Meanwhile, the confidence interval criterion evaluates the confidence intervals for the predictions of the hybrid model [95]. Other authors have proposed that the confidence interval of the hybrid model can be estimated using Bayesian optimization, resampling the model's prediction using Markov Chain Monte Carlo simulations [96].

3.6 | Hybrid Modeling Requirements for Data

Data is essential for discerning the structure and parameters of the hybrid model, and principally all properties of the model (such as prediction quality and extrapolation capabilities) heavily rely both on the quality and quantity of the data [10, 12, 24]. Data is necessary to develop any hybrid model. If there is no previous data, it must be generated, with a systematic exploration of the design space (e.g., using Design of Experiments for a system or a benchmark model). Different techniques can be applied, such as factorial design, sequential design, optimal design, and model-based design methodologies [97]. A valuable alternative is the so-called "Intensified Design of Experiments," where intra-experimental shifts to key parameters are performed to capture the dynamics of the process [59, 98, 99]. This methodology has been applied specifically to develop hybrid models, yielding great predictive power with a reduced number of experiments in an early stage of model development. Intensified design of experiments has been paired with hybrid models to transfer knowledge from lab scale to pilot scale in CHO cultures [100], reducing the number of experiments by as much as 50% while retaining the same levels of accuracy in monoclonal CHO line producing immunoglobulin G4 monoclonal antibody [101, 102], describing the design space with similar accuracy as a full-factorial design of experiments despite having less than 33%

of the experiments in an *E. coli* culture for recombinant protein production [103].

Regardless of the experimental design, data should be cleaned and properly pretreated before using it for hybrid modeling purposes. For example, small variances in feeding concentration can cause big errors in the estimation of substrate concentrations [57]. Hybrid model performance is affected by higher levels of noise, increasing the variance of the identified model parameters [104, 105]. Noise from data can be extracted using techniques such as outlier removal and signal filtering [106] or using surrogate models for data augmentation to increase model stability in predictions [23, 107, 108].

3.7 | Hybrid Modeling Applications

A classification according to the use given to the hybrid model has been proposed, providing in-depth examples for each category [12]. Based on that classification, the main uses for hybrid modeling in bioprocesses are shown in Table 2.

Besides the previous applications, hybrid models have gained momentum for the development of digital shadows and digital twins, to accelerate Industry 4.0 and digitalization in chemical processes [16, 119].

In the following sections, insights regarding hybrid modeling of bioprocesses are presented. Due to the vast number of references

analyzed, only a select number of them will be included in the text as examples. The database of papers analyzed will be made available on request.

4 | Hybrid Modeling Implementation

Recently, a comprehensive bibliometric analysis [52] spanning 360 contributions revealed that the hybrid modeling of bioprocesses is a growing research area, with an annual growth rate of 9% since its inception in 1992. Recent advancements, particularly post-2017, underscore the impact of heightened awareness surrounding artificial intelligence and the integration of next-generation ML techniques into hybrid models. Key themes driving this field include “hybrid model,” “neural networks,” and applications ranging from fermentation to biopharmaceutical manufacturing. A keyword network analysis identified three central research clusters: process control applications, computational techniques, and the broader integration of bioprocessing terminology with ML concepts.

This review builds on these findings, delving into quantifying trends, innovations, and challenges in hybrid modeling for bioprocesses, setting the stage for a comprehensive examination of the current state of the field to identify trends such as the types of hybrid modeling on bioprocesses, the preferred design patterns, the most used nonparametric techniques, the scale of the experimental data for developing the models, the mode of operation of the experiments, software used for building the models, as well as

TABLE 2 | Main uses for hybrid modeling and some applications.

Type of use	Case study
Process Monitoring	- Predicting mixed liquor suspended solids, chemical oxygen demand, and nitrate concentration in the permeate of a membrane bioreactor for wastewater treatment [109] - Monitoring penicillin G acylase production by <i>Bacillus megaterium</i> using principal component analysis for detecting deviations from the desired trajectory [110].
Process Control	- Applying Support Vector Regression and Universal Kalman Filters in a therapeutic protein production using <i>E. coli</i> [111] - Using online measurements such as pH, temperature and dissolved oxygen to control a target metabolite in a <i>Bordetella pertussis</i> cultivation [112]
Process Optimization	- Using ANNs in conjunction with the material balances to optimize the feeding strategy of a fed-batch bioreactor to maximize the yield of a protein [113], - Using ANNs and particle swarm optimization to determine the optimal conditions that maximize ethanol production in a fermentation by <i>S. cerevisiae</i> grown using cashew apple juice [114].
Scale-up	A hybrid model was developed using data from 300 mL flasks and applied it to describe the process in a 15 L bioreactor [100].
Analysis of model-plant mismatch	- Determining the deviations and correcting the predictions of pH, chemical oxygen demand and other variables in a wastewater treatment plant [115], - Monitoring the soluble chemical oxygen demand for improving the nitrogen removal efficiency in a wastewater treatment plant [116].
Model Transfer	- The use of meta-learning and one-hot encoding to transfer knowledge from one hybrid model to another process reducing test error when compared to a model trained only in the new experiments [117] - Using process data in one experimental condition to transfer the knowledge to another set of conditions, improving the behavior of the model [118].

current challenges and future potential. This section highlights selected examples to illustrate the main macro trends in the field, rather than providing a detailed account of all contributions.

Section 4.1 presents an overview of the most used nonparametric substructures for hybrid models. Section 4.2 presents the types of processes that have been successfully modeled using hybrid semi-parametric approaches with the most used microorganisms in those processes. Section 4.3 presents the type, scale of the data and mode of operation of the processes used for hybrid modeling. Section 4.4 presents some computational details regarding the implementation of hybrid models, and finally Section 4.5 presents some recent trends.

4.1 | Data-Driven Substructures Model Approaches

Figure 4 shows the distribution of the data-driven techniques used for hybrid modeling. A total of 374 different hybrid model implementations were found in the literature. It's important to note that there are more individual implementations than works analyzed because one work can have more than one implementation. ANNs dominate as the preferred black box component in hybrid modeling, accounting for 56.2% of the total modeling

approaches. Inside the ANN set, 67% of the authors that use ANNs do not specify the type of ANN they use, for example, using ANNs to develop hybrid models for CHO cultivation from shake flask cultivations and yielding similar accuracy to the mechanistic model despite using only 4 of 33 experiments for training [120], or to describe a *B. subtilis* fermentation to maximize the spores produced [121]. The remaining authors have used the following types: multi-layered perceptron (MLP), which are 13% of the applications and have been used to describe the reaction rates in perfusion cultures of mammalian cells, reducing the elementary fluxes modeled from 22,563 to 5 [60], or to predict a CHO cultivation process by replacing reaction rates with MLPs [122], or to model the enzymatic kinetics for producing galactooligosaccharides using β -galactosidase from *Kluyveromyces lactis* [123]. Deep Neural Networks (DNN) comprise 4% of the total ANN applications and have been used to describe processes such as the production of glycoproteins in a CHO culture, resulting in an error 12% lower when compared to the shallow counterparts despite an increase in computation time of 31% [124], modeling the *E. coli* threonine synthesis [125], or modeling real industrial fermentations with a 75% reduction in RMSE [126]. Feedforward Neural Networks (FNN) which account for 3% of the total ANN applications, have been used to predict the gas composition of a biomass gasification process [127] or to predict the insulinase

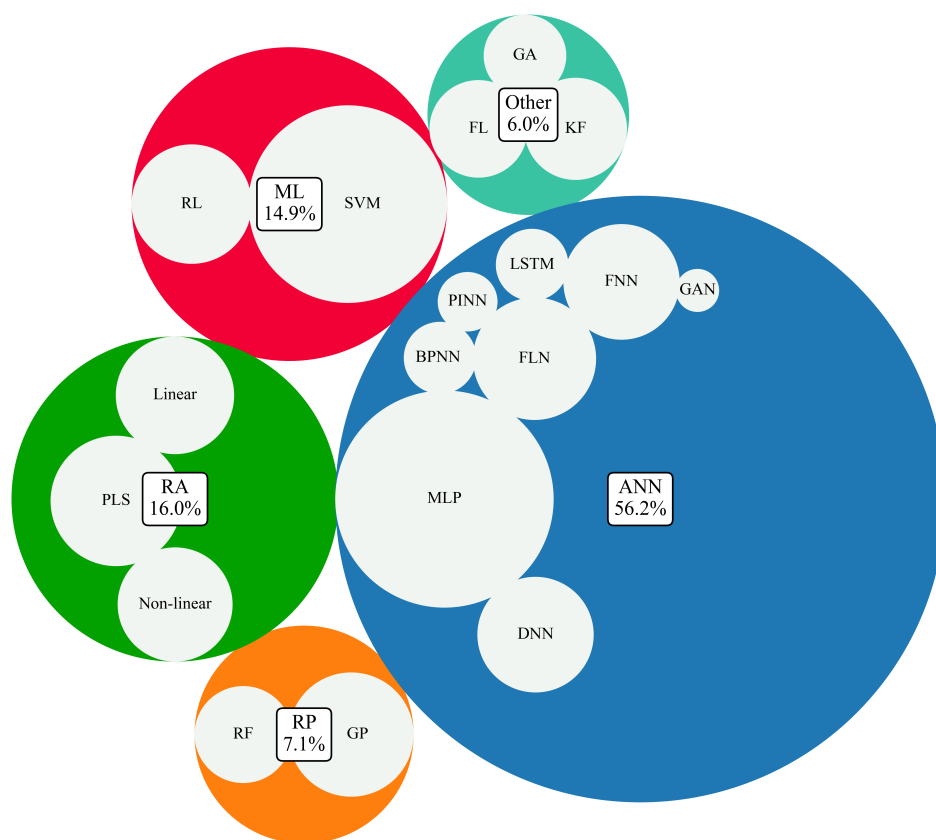


FIGURE 4 | Distribution of black box models' categories and subcategories used by authors when implementing hybrid Models in bioprocesses. Bubble size indicates number of occurrences in the database. ANN, artificial neural networks; BPNN, backpropagation neural networks; DNN, deep neural networks; FL, fuzzy logic; FNN, feedforward neural networks; GA, genetic algorithms; GAN, generative adversarial neural networks; GP, Gaussian Processes; KF, Kalman Filter; LSTM, long short-term memory neural networks; MARS, multi-adaptive regression splines; ML, machine learning techniques; MLP, multi-layered perceptron; PINNs, physics-informed neural networks; PLS, partial least squares; RA, regression analysis; RF, random forests; RL, reinforcement learning; RP, random processes; SVMs, support vector machines.

production in a fed-batch bioreactor using *Kluyveromyces marxianus*, where hybrid models performed up to 12% better than the parametric models [128]. Functional Link Neural Networks (FLN) comprise 3% of the ANNs subset and have been used to predict the behavior of a series of industrial ethanol fermenters considering the effects of recycling and temperature changes [129, 130]. PINNs are an emerging implementation with 2% of the total ANN uses in hybrid modeling of bioprocesses, for example, predicting β -carotene production using *Saccharomyces cerevisiae* and reducing MSE by 80% when compared with a traditional parametric model [131]. Backpropagation Neural Networks (BPNN) which represent 1% of the applications have been implemented to model the production of sodium gluconate in a fermentation by *Aspergillus niger*, with the hybrid model yielding errors up to 87% lower than the parametric mechanistic model [132], or to identify the metabolic fluxes in a recombinant Baby Hamster Kidney culture [133]. Long Short-Term Memory (LSTM) neural networks comprise 1% and have been used for predicting the pH of cream cheese fermentation at both lab and industrial scales [134] or for predicting the ethanol concentration in industrial fermenters aided by the use of Generative Adversarial Neural Networks (GAN), with 1% of the total uses of ANNs, for replacing missing datapoints of yeast [135].

In Figure 4, the Regression Analysis (RA) category refers to regression techniques that are not random processes nor ML nor ANNs, such as series expansions, polynomial expressions or time-series models like ARMAX. This category is the second biggest approach, with 16% of all the applications having a vast array of techniques at their disposal. In this category, Partial Least Squares (PLS) is the most used technique (36% of the occurrences) and has been used to describe the formation of recombinant products reducing errors by as much as three orders of magnitude when compared to traditional metabolic flux analysis [136], or to describe the mass transfer coefficient in a MBR for WWT, reducing the RMSE by 52% [137]. Linear techniques follow with 32% of the applications in this subcategory. This subcategory refers to techniques such as multivariate curve resolution coupled with mass conservation equations to describe CHO cell cultivation processes [100, 138]. Also, polynomial approaches for describing lutein production by microalgae have errors up to 5% lower than the mechanistic approach [139]. “Non-linear” refers to various techniques used in bioprocess modeling to improve predictive accuracy. One example is non-linear PLS, which has been applied to predict concentrations based on spectral data, yielding lower errors than traditional chemometric tools [140]. Another application involves modeling the specific reaction rates of a *Bordetella pertussis* culture by integrating both online and offline measurements, enhancing process understanding and control [112]. Similarly, multi-adaptive regression splines (MARS) have been employed in a simulated penicillin fermentation process, demonstrating superior predictive performance with mean errors 15 times lower than the parametric approach [141].

ML techniques comprise 15% of the applications. SVMs account for 47% of the applications in this subcategory and have been used in fluxomic for identifying genes to overexpress and other genes to suppress resulting in an enhanced ethanol production by 10% [142], or to optimize a nosiheptide fed-batch fermentation increasing yields by as much as 26.8% [143] or used to predict the

mycelial concentration in a penicillin fermentation [144]. Reinforcement Learning (RL) accounts for 16% of the applications in this subcategory, having been used to enhance the yield prediction in a *Yarrowia lipolytica* fermentation process [145]. Other ML applications used are Extreme Gradient Boosting (XGB) for predicting viable cell density and glucose consumption in a CHO culture [146] or for predicting the rate constants in the reaction between alkyl esters and hydrogen atoms in biodiesel combustion processes [147].

Random Processes (RP) comprise 7.1% of the hybrid model's applications, with examples such as GPs that account for 42% of this subcategory and have been used to predict the effluent composition of a full-scale WWT plant [148], or to predict an astaxanthin production system reducing MAPE by two orders of magnitude [149]. RF that comprise 18% of this subcategory have been used for predicting the production of a therapeutic protein in a CHO culture [146] or to predict the chemical oxygen demand and nitrate concentrations in a WTT plant with errors 8% lower than traditional activated sludge models [150].

The “Other” category accounts for 6% of the applications, mostly composed of Kalman Filter (KF) which is 32% of this subcategory and has been applied to reconstruct the experimental data from an *E. coli* fermentation [151], or to predict neuron activity in mammalian cells [152]. FL comprises 25% of the applications in this subcategory and has been used to predict the different stages of a fermentation thus adapting the model for each phase [153–155]. Genetic Algorithms (GA) are 25% of the approaches in this subcategory and have been adapted for optimizing hybrid models that describe *E. coli* fermentations [156, 157] or to identify and optimize the structure and parameters of chemical reaction networks [158].

4.2 | Processes Where Hybrid Models Have Been Applied

The type of application for which the modeling objective of the hybrid models is shown in the innermost ring of Figure 5. One hundred fifty-five different implementations were found in the literature. Most of the documents are related to Metabolite Production, processes that use hybrid models aiming to describe the concentration or behavior of a substance produced by a microorganism. The second biggest category is Cell Cultivation, where the goal of the application is to produce biomass of a microorganism. WWT is the third category, where hybrid models have been used to describe biological treatments. The “other” label accounts for processes like biosorption [159], transesterification [160], membrane separations of fermentation broths or other bio-based processes [161, 162], among others.

The second ring shows how authors describe their applications. The Cell Cultivation section shows that all of them are naturally inclined to describe their processes as biomass growth. In the Metabolite Production category around half of the applications are for fermentations, followed by Protein Production and the description of Enzyme kinetics and activity of different bioprocesses, such as cellobiose using glucosidase [163], or the proteolysis of whey by alcalase [164]. In the WWT application, hybrid models have been used to describe activated sludge reactors (AS)

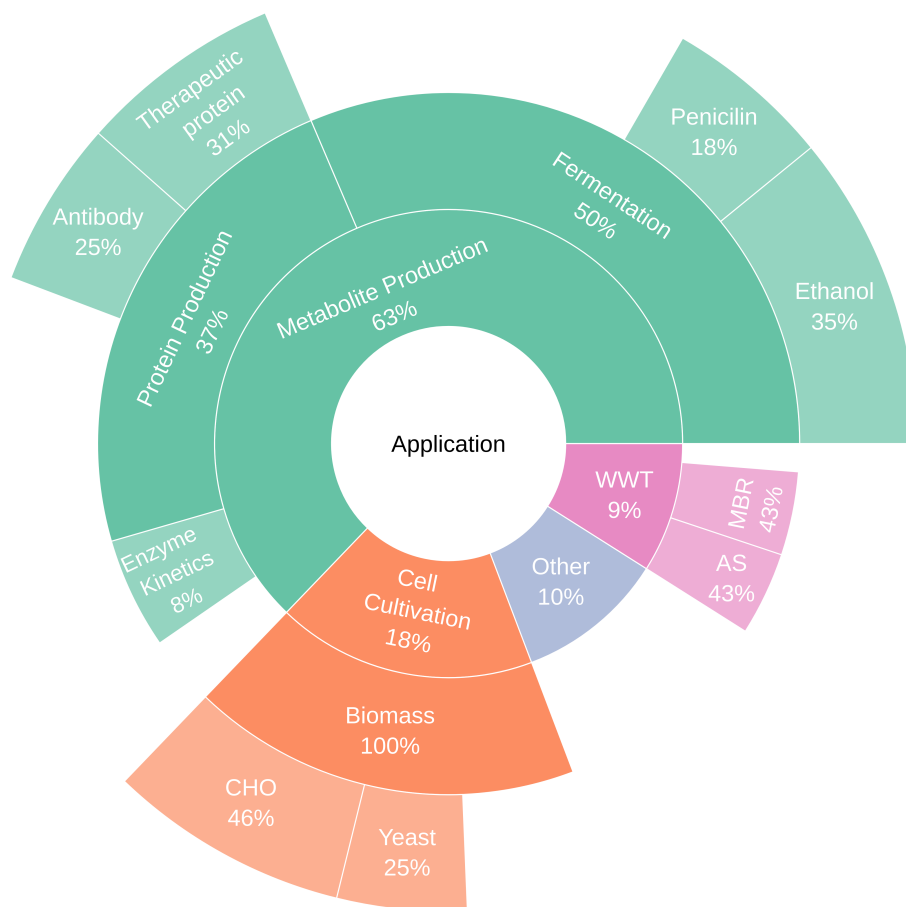


FIGURE 5 | Type of bioprocess hybrid models have been applied to, categorized by the main target of production: Metabolite Production, Cell cultivation for obtaining biomass, wastewater treatment (WWT), and other processes. CHO represents Chinese Hamster Ovary. AS stands for Activated Sludge reactor, and MBR stands for Membrane Bioreactor.

using synthetic data generated from benchmark models [75], ANNs for predicting reaction rates in a simulated phosphorous removal plant [165]. Also, GPs are used to fill in data gaps, thus, enhancing model prediction in a parallel architecture [148], predicting the reaction rates of nitrification and denitrification in a series of tanks in an industrial sludge WWT plant [166], the predictions of pH, chemical oxygen demand and other variables in a WTT plant [115]. Besides, for WWT there are also MBRs, where the models have been used to predict biochemical oxygen demand, chemical oxygen demand, total nitrogen, and total phosphorus in a municipal WWT plant [167], suspended solids, chemical oxygen demand, and nitrite and nitrate concentration in the permeate of a MBR for WTT [109]. Also, there are applications to predict the counterion mass transfer in a membrane-supported bioreactor designed for the removal of nitrate and perchlorate from a synthetic contaminated drinking water stream [137], or to predict in real-time the phosphate concentration from municipal wastewater [168].

The outermost ring shows the products of interest most studied. The most common products for fermentation studies accounting for more than half of the applications are bio-ethanol production [135, 169–178] and penicillin [15, 61, 144, 153, 179, 180]. Other applications include the production of xanthan gum

by *Xanthomonas campestris* [181], a fermentation using *Streptomyces actuosus* for producing nosiheptide [182], the production of linolenic acid [183], astaxanthin production [149], the production of sorbitol by *Zymomonas mobilis* [184], lactic acid production using *Streptococcus* [77], the production of pharmaceuticals such as paclitaxel [185], and a *Pseudomonas ovalis* fermentation for producing gluconic acid [186].

For “Protein production,” most studies with around 57% of the documents focus on antibodies [101, 102, 187–192] and therapeutic proteins [98, 193–196]. Other proteins include streptokinase production [197], threonine production [125], and lipase production [198].

In “Cell Cultivation” applications, the field is dominated by CHO cells [122, 199–201] and *Saccharomyces cerevisiae* (displayed as “yeast” in Figure 5) [78, 131, 170, 202]. Other cases include the cultivation of *Chlorella vulgaris* [203], cultivation of *Bacillus* [204, 205].

Figure 6 presents the percentage correlation between the black-box techniques used for developing hybrid models and the type of processes they have been applied to, where each percentage represents a fraction of the total applications. ANNs dominate

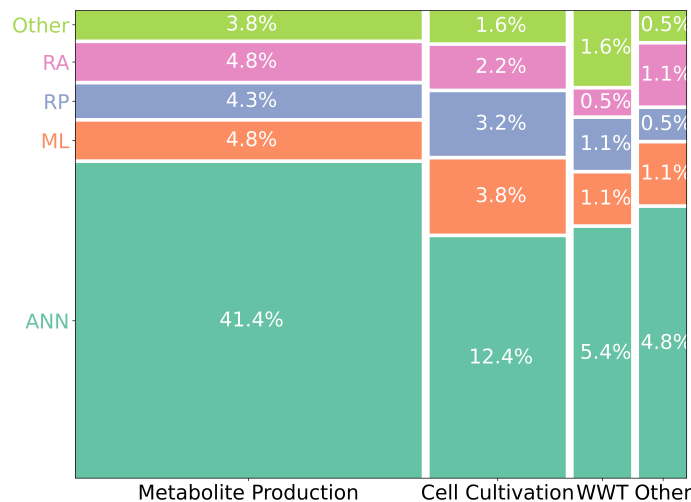


FIGURE 6 | Cross-reference of types of black box used for building hybrid models and the target processes. ANN, artificial neural networks; ML, machine learning; RA, regression analysis; RP, random processes.

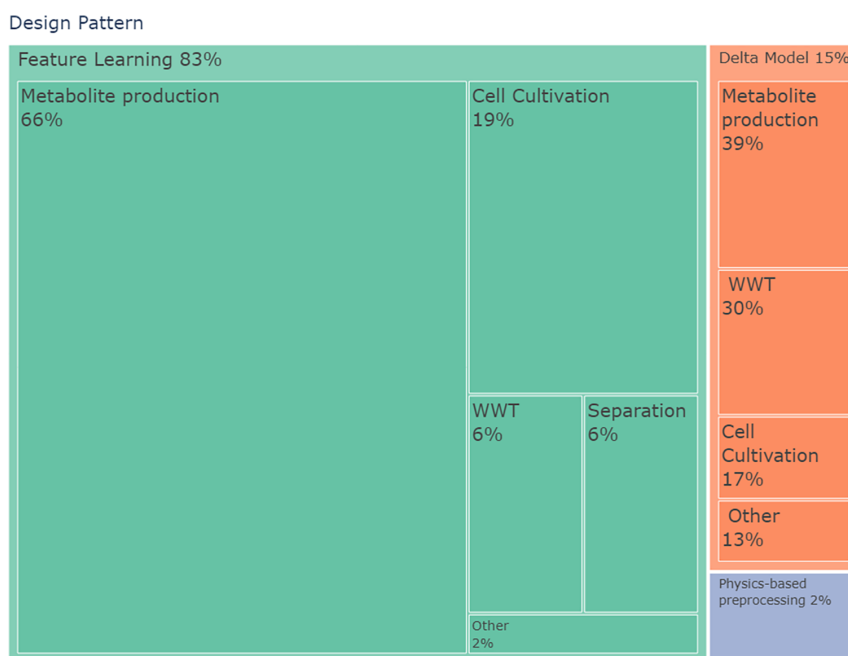


FIGURE 7 | Distribution of design patterns by type of process it has been applied to.

all the different applications, with a wide margin over the different ML techniques. In Metabolite Production applications, ANNs comprise over 41% of the total applications, more than doubling all the other different black-box techniques. However, when moving to the other categories, the margin between ANNs and other types of technologies diminishes. ML techniques are the second most used black-box component overall, followed by RP, RA and other techniques. It is interesting to note that for “Metabolite Production” applications, RA is the second most used technique, but falls behind RP and ML. Other techniques are the least used, except for WWT where it becomes the second most used.

One important aspect to consider is that adding the percentages of each type of application in Figure 6 does not match with the percentages given in Figure 5 due to authors exploring the use

of different black-box components and combinations for their hybrid models. If a contribution uses two different black-box techniques for the same application, these are counted as separate instances (hence “double counting” some studies), which explains why “Metabolite Production” has a lower percentage and the other categories have higher percentages when compared to Figure 5.

The most common architecture used is the “feature-learning” design pattern with 83% of the applications, followed by “delta model” implementations and “physics-based preprocessing” with 15% and 2% respectively (Figure 7). Some authors use a mixture of both parallel and serial architectures [144, 148, 206]. In all the design patterns, metabolite production comprises the

bulk of the applications, although inside the “delta model” implementations, the margin is thinner than in “feature learning” applications. It is important to note that Cell Cultivation processes have the same share in “feature learning” implementations than “delta model”. However, this is not the case for WWT, signaling that authors prefer “delta model” implementations when dealing with these case studies. This might be due to higher uncertainty and less mechanistic knowledge available when modeling the complex processes of WWT. Many of the WWT applications use hybrid models to predict contaminant concentrations [75, 148] or physicochemical parameters like BOD, pH, and COD [207, 208].

Regarding the outputs of the nonparametric components of hybrid models, common predictions are kinetic or transport parameters with 55% of the contributions, followed by 19% that predict reaction rates, 11% directly predict concentrations, and 15% predict other quantities such as pH, fluxes, or volumetric flows. “Delta model” architectures usually model the residuals or the mismatch [209–211].

Figure 8 shows the distribution of the microorganisms and enzymes used in processes where hybrid models were implemented (no matter if the goal was cell cultivation or metabolite production). *Saccharomyces cerevisiae* is the most studied microorganism with the largest number of applications coming from ethanol production or cell cultivation [67, 78, 131, 135, 170, 173, 202, 212–216]. Then, *E. coli* which has been mainly used for antibody and protein production [99, 151, 194, 217–220], has also been used for producing antivenom [221]. CHO has been mainly used in therapeutic production processes [51, 100, 122, 146, 199–201], with the three accounting for more than half of all applications. *Pichia pastoris* has been studied mainly for protein production [188, 195, 222], whereas *Zymomonas mobilis* has been used to describe sorbitol production [184] and ethanol production [174, 175]. The label “Bacteria” refers to unspecified bacteria, usually associated with mixed cultures in WWT [223, 224], but also *Enterobacter* A47 to produce exopolysaccharides [225], or the production of bacterial cellobiose by *Acetobacter xylinum* [226]. The category “Other” encompasses 32 different microorganisms and includes MA-104 (epithelial cells from the kidney of an African green monkey) for vaccine development [227], *Kluyveromyces lactis* for producing galactooligosaccharides [123], *Xanthophyllomyces dendrorhous* for producing astaxanthin [149], mixed microbial cultures for producing polyhydroxyalkanoates [228], *Yarrowia lipolytica* for citrate production [145], *Cunninghamella echinulata* for linolenic acid production [183], *Pseudomonas putida* used in the production of polyhydroxyalkanoates [229], *Aspergillus niger* in a sodium gluconate fermentation [132], *Ralstonia eutropha* in the production of poly- β -hydroxybutyrate [230], *Candida rugosa* in the production of lipase [198] among others.

4.3 | Type of Data and Mode of Operation

Most of the hybrid models’ applications use experimental data for their development (80% of the total). The remaining 20% of the contributions use simulated data, usually derived from other mechanistic models such as the production of gluconic acid by *Pseudomonas ovalis* [231], describing the interactions between

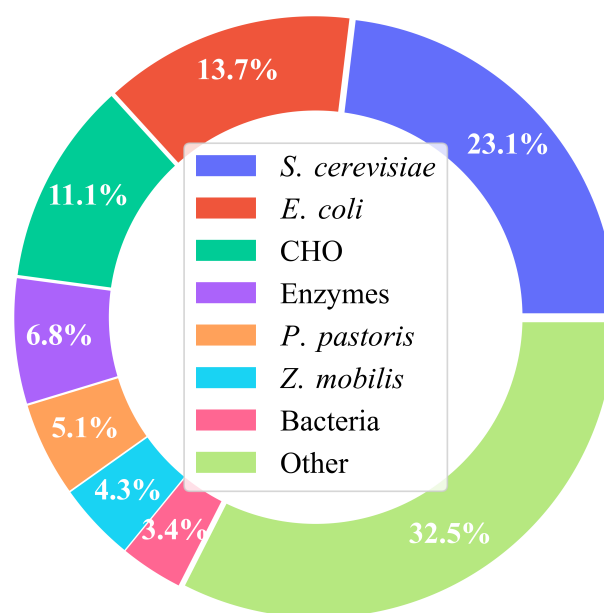


FIGURE 8 | Most used microorganisms in the works for developing hybrid models.

Chlorella vulgaris and *Stutzerimonas stutzeri* for chlorophyll accumulation [232], applying inverse PINNs to model complex reaction networks [74], *E. coli* for recombinant protein production [233], adapting symbolic regression to infer the structure and parameters when building hybrid models [234] or unnamed biological systems [235, 236]. It is important to note that hybrid models trained on simulated data can present a high level of bias because of the propagation error of the models used for producing the data.

Figure 9 shows the scale of the experiments and the operation mode used to generate the calibration data. On the left, the scale of the experimental data used for developing hybrid models is compared to the type of application (as shown in the inner ring of Figure 5). The most common scale is the laboratory for all applications. For pilot scale, cell cultivation applications include BHK-21 [133] and an undisclosed microorganism [79], Metabolite production applications at the pilot scale include cellulose production using *Acetobacter xylinus* [226], optimization and control of a *Bacillus thuringiensis* fermentation [237], a methyl methacrylate polymerization reactor, the production of a foreign protein by an undisclosed bacteria [223], several studies using *Pichia pastoris* for antibody production [188, 222, 238], *Aspergillus niger* fermentations for glycerol production [119] and sodium gluconate [132], using *Streptomyces actuosus* for nosiheptide production [182], diverse cases of *E. coli* for protein production [192, 219], ethanol production using *S. cerevisiae* [64, 114], and therapeutic protein production using CHO [100]. Pilot scale WWT data have been used to build soft sensors for modeling the phosphorous removal in a biofilm reactor [116], prediction of effluent quality in an activated sludge reactor, and control of critical parameters in an alternating aerobic-anoxic WWT plant [166]. At industrial scales, there are several applications for cell cultivation of *S. cerevisiae* [78, 239] and a CHO cultivation process [240]. Inside metabolite production applications at the industrial scale, ethanol production using *S. cerevisiae* [58, 129, 130, 241],

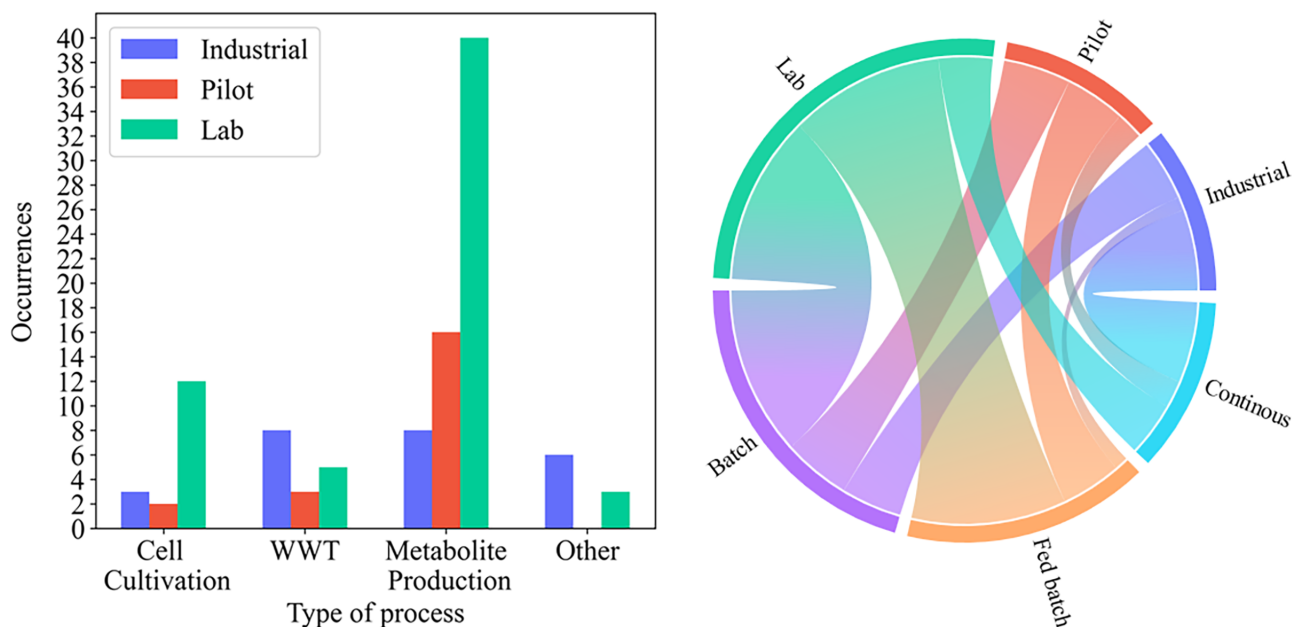


FIGURE 9 | Left: Number of studies showing the scale of the experimental data used for developing the hybrid models according to the type of application. Right: Relationships between the scale of the data used for developing hybrid models and the mode of operation.

an undisclosed industrial fermentation modeled using deep neural networks [126], pH prediction at an industrial cream cheese fermentation [134], erythromycin fermentation [242], and *E. coli* for therapeutic protein production [59, 99]. Other applications at industrial scale include ultrafiltration of fermentation broth [162], chromatographic processes for protein capture [243], and WWT [66, 109, 115, 148].

In Figure 9 on the right, the cross-reference of scale and mode of operation is shown. Batch experiments are the most common accounting for 44.6% of the category; fed batch and continuous systems trail behind with 33.7% and 20.7% of the applications respectively, while around 1% of the contributions do not report the scale of their experiments. Industrial datasets comprise 21.7% of the contributions, with pilot scale datasets accounting for 25.3% and laboratory scale including 53% of the contributions. Industrial batch applications account for 9.7% of the contributions, with continuous systems accounting for 9.6% and fed batch systems including 2.4% of the contributions. At the pilot scale, batch reactors comprise 12% of the contributions, with continuous and fed-batch reactors accounting for 3.6% and 9.6% respectively. Finally, at the lab scale, continuous reactors comprise 8.4% of the contributions, with fed-batch accounting for 21.7% and batch representing 22.9% of the contributions.

4.4 | Computational Aspects

Only around 40.3% of the documents report on the software used for implementing the hybrid models. Figure 10 at the left shows the distribution of software, whereas on the right a cross-reference of the software and the nonparametric structures is shown as described in Figure 4. ANNs are the preferred nonparametric structure used in every software, followed by Regression techniques. Genetic algorithms were found only in MATLAB applications whereas Python presents several ML techniques,

primarily developed in TensorFlow, PyTorch and Sci-kit learn packages.

Most of the hybrid models have been implemented in MATLAB [17, 59, 99, 164, 244–246], with Python rising in recent years [122, 147, 209, 220]. Some applications have been developed in software like C# [241], Julia [131], Statistica [163, 181], or FORTRAN [74, 247].

Figure 11 shows the performance indicators for evaluating hybrid model's fit. The most used performance indicator is the family of the MSE, which encompasses 45% of the occurrences (244 performance indicators used to evaluate models). Authors have used MSE in various forms, such as ROOT MSE (RMSE), normalized MSE (NMSE), RMSE of predicted values (RMSEP), weighted MSE (WMSE), and base MSE among others. The second most used indicator is r^2 (Coefficient of determination), appearing in 25% of the applications, even though it may not be suitable for many non-linear problems. The Information Criteria (IC) such as Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC) are very useful for determining the structure of the nonparametric sub model [11, 57]. Errors such as MAPE, MAE, mean relative error (MRE), mean average error (MAER), residual standard error (RSE), and mean relative percentage error (MRPE) make up almost 11% of the instances. The “Other” category is dominated by sum of squared errors (SSE) and residual standard deviation (RSD).

4.5 | Recent Trends

Recent applications of hybrid modeling include the work of [248], where the authors use Bayesian inference applied to a Neural Network designed to predict both the reaction rates and stoichiometry from simulated data. They compare their results to a

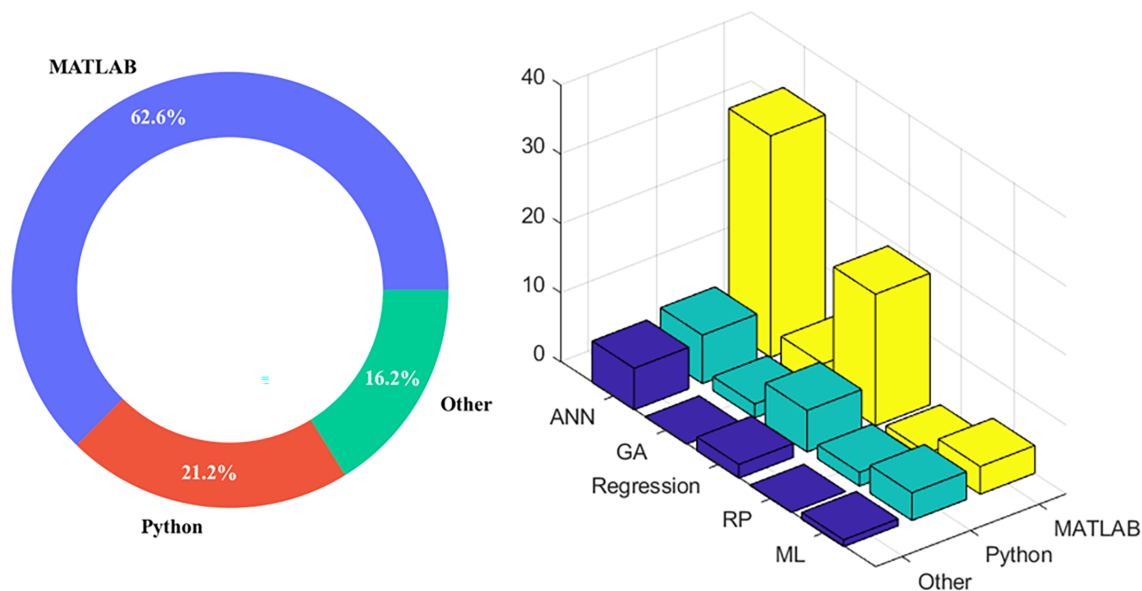


FIGURE 10 | Left: Software distribution. Right: Type of nonparametric structure by software.

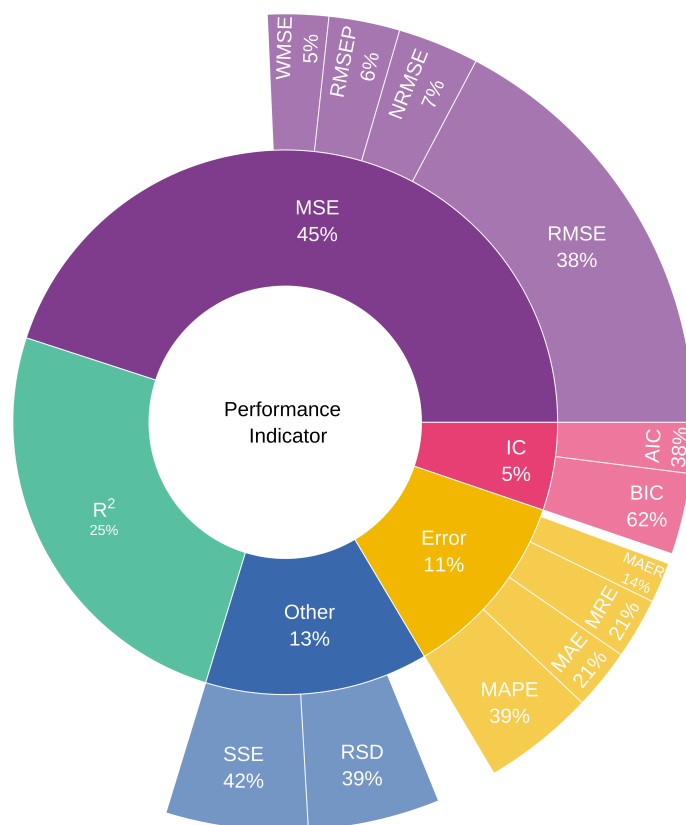


FIGURE 11 | Performance indicators used to evaluate the hybrid model's capabilities. AIC, Akaike Information Criterion; BIC, Bayesian Information Criterion; IC, Information Criteria; MAE, mean absolute error; MAER, mean average error; MAPE, mean absolute percentage error; MRE, mean relative error; MRPE, Mean relative percentage error; MSE, mean squared error; NMSE, normalized MSE; RMSE, root MSE; RMSEP, of predicted values; RSD, residual standard deviation; RSE, residual standard error; SSE, sum of squared errors; WMSE, weighted MSE.

pure data-driven technique, obtaining narrower confidence intervals and better prediction capabilities. The production of threonine applying Deep Learning techniques for the nonparametric model, obtains low errors when compared to the synthetic ANN

training dataset [125]. Deep Learning was also applied to model a 50 L *Pichia pastoris* cultivation, increasing prediction accuracy by 18.4% and reducing CPU time by 43.4% when compared to a shallow ANN [238]. Hybrid models have proven to perform better

for antibody production than traditional parametric models [24, 102, 249]. A hybrid model was developed using data from 300 mL flasks and applied to describe the process in a 15 L bioreactor, with the error being in the same range as the analytical error and that of a traditional regression model [100].

A 100 kilogallons fermenter has been recently studied [126]. The authors carry out a sensitivity analysis of the traditional kinetic model to better understand which factors affect the output the most. The identified kinetic parameters are selected as candidates to be modeled using a Deep Neural Network. Of the candidates, a clustering analysis is done to identify those that vary with time, and the ones that have time dependence were selected as the parameters to be modeled by the nonparametric structure. The resultant hybrid model outperforms the traditional kinetic model, reducing the RMSE by a factor of 10.

Some authors have used hybrid models to infer the structure of the mechanistic model [105]. A hybrid model using RL to regress the kinetic parameters for two proposed parametric structures was developed. The used framework correctly identified that a Monod-like kinetic structure was wrong because the parameter values were 0. In the same study, the authors trained a hybrid model on synthetic data (generated using the identified models) which predicted time-variant kinetic parameters, achieving a MAPE lower than 5% for an experimental batch run. Finally, they investigated the effect of noise in the hybrid model's prediction. The MAPE deteriorated to 17% when 30% noise was added, indicating that the proposed framework can deal appropriately with noisy data.

β -carotene production using *Saccharomyces cerevisiae* has been studied [131]. The authors used a Universal Differential Equation (UDE) to develop their hybrid model. In the UDE approach, a neural ordinary differential (an ordinary differential equation that is described by a neural network) equation that approximates the derivatives of the previously unknown dynamics of the batch fermentation process is integrated with the ordinary differential equations (ODEs) of its kinetic model to give a hybrid model with superior accuracy. The UDE model is compared with a traditional kinetic model, resulting in better predictions and lower MSE. Their work shows that neural ODEs can be integrated into hybrid models to describe kinetic parameters and their changes over time.

The level of “grayness” in hybrid models was investigated by [183] in a *Cunninghamella echinulata* fermentation. Experiments were carried out at a 1 L fermentation and the models were tested in a 5 L bioreactor. Three hybrid models were developed, using GPs as a nonparametric structure, with an increasing degree of mechanistic knowledge in the kinetic expression. The model with the most mechanistic knowledge performed worse, indicating a high risk of inducing bias that hinders the model's performance. The hybrid models showed similar performance in the 1 and the 5 L training datasets. The model with intermediate mechanistic knowledge performed better, with lower uncertainty and error. Introducing limited mechanistic knowledge resulted in overfitting issues. Their results show that incorporating kinetic knowledge that is not well understood can harm a hybrid model's performance. A similar study was conducted in a CHO bioreactor [122]. Authors investigated several parametric and

nonparametric techniques. However, they also investigated the “grayness” of hybrid models. Different hybrid models replacing one flux and up to five kinetic parameters predicted by the nonparametric technique were tested against an established parametric model. The hybrid models reproduced the design space, although on some edges the predictions were flatter than the ones given by the parametric model.

Digital shadows, which are in silico representations of physical systems that capture and aggregate data from their real-world counterparts and have been built as a decision-making tool, present themselves as an emerging use case for hybrid models [250]. Recently, authors developed a digital shadow by coupling a mechanistic model with a purely data-driven soft sensor for estimating glucose and glycerol concentrations [119]. They studied a pilot (volumes of 550 and 2500 L) *Aspergillus niger* fermentation. The model was trained in 55 batch experiments with 4 additional calibration runs. To study scalability, 8 tests at the lab scale were performed. For the development of the data-driven soft sensor, a variety of models were tested, including ANNs, SVM, RF, Partial Least Squares, XGB, and LGBM. Their implementation achieved a normalized residual MSE of 2% with an r^2 of 0.98. The model predicted biomass, glucose, and glycerol concentrations accurately for the three scales, indicating that hybrid models could be developed in smaller scales and later implemented in bigger scales, and still retain high prediction accuracy.

Hybrid models have been recently used to accelerate the design and optimization of monoclonal antibody production [191]. The methodology used in silico experimental runs with a model derived by other authors [251]. The in silico experiments were used to train a hybrid model that follows a “feature learning” design pattern that uses an ANN as a nonparametric component. After the hybrid model training, a genetic algorithm was used to optimize the feeding schedule that maximizes the antibody titer at harvest, and after identifying the optimal conditions, another in silico experiment was run to assess the reliability of the predicted values. This strategy was compared to a traditional response surface methodology optimization strategy. The hybrid model correctly identified the optimal feeding strategy within 1% of the real strategy and showed a 34.9% increase in antibody titer when compared to training data and a 2.8% higher concentration than the optimal results from traditional response surface methodology. However, the predicted titer of the hybrid model was 18.6% lower than the real one, because the authors were extrapolating the hybrid model outside of its training region. Adding one more in silico experiment that covered the extrapolation region reduced the prediction error to 8.7%.

LSTMs have been recently applied to model a downstream separation in an industrial biorefinery manufacturing 1,4-bio-butanediol [162]. The authors utilized a PLS model for estimating trans-membrane pressure, which was then coupled with a mechanistic knowledge model that follows Darcy's equation, in a “feature learning” design pattern. The mechanistic knowledge model estimated the membrane resistances, and the hybrid model was deployed to predict short-term and long-term fouling of membranes, achieving r^2 higher than 0.9. The authors claim that the model can identify irreversible and reversible

fouling, allowing for better analysis of performance and more in-depth diagnostics.

Hybrid models have been used in genome-scale metabolic modeling to optimize ethanol production using *S. cerevisiae* [142]. Authors propose a hybrid model-driven platform that integrates flux balance analysis and several ML techniques, such as SVM, MLPs, RFs, and several linear regression techniques. The study highlights the importance of combining metabolic modeling and ML to unravel the complex relationships between genotype, phenotype, and the environment. The hybrid model identified key genetic modifications, including the overexpression of six target genes and the knockout of seven target genes, which significantly improved ethanol yield. Experiments showed that knocking out succinate dehydrogenase genes increased ethanol production by 6%–10% compared to the wild-type strain. Further dual-gene deletion experiments targeting glycerol-3-phosphate dehydrogenase and succinate dehydrogenase resulted in even higher ethanol yields, with improvements of up to 27.9%.

Recently, dynamic modeling encompassing first principles and recurrent neural networks (RNN-LSTM) has been developed to enhance the sporulation process of *Bacillus licheniformis* BF-002 [204]. This innovative approach aimed to improve spore production by optimizing cultivation conditions, including carbon/nitrogen source feeding rates, pH, temperature, and agitation speed. The researchers cultivated the strain in a 5 L fermenter. The hybrid model was implemented in a “feature learning” design pattern: a traditional dynamic model that described the temporal variations of vegetative cell quantity and nutrient concentrations during growth and sporulation, and an RNN-LSTM model that predicted kinetic parameters dynamically influenced by environmental conditions. The hybrid model achieved a reduction of MRE of up to 82% in the number of vegetative cell quantities and 22% in the spore quantities when compared to a parametric model. The hybrid model was used with genetic algorithms to find the optimal feeding conditions that maximize spore production. This approach resulted in a 35.21% increase in sporulation, highlighting its effectiveness in optimizing cultivation dynamics.

Transfer learning approaches have been recently implemented to take advantage of existing knowledge of one CHO process to another [146]. Authors used MLPs, RFs, XGBoost to predict kinetic parameters in a “feature learning” design pattern. There were two sets of experiments that produced different antibodies. One set was used for training and calibrating the hybrid models (dataset A) and the other was used for testing the transfer learning capabilities (dataset B). To correctly identify the adequate inputs to the nonparametric techniques, a series of correlation analyses and multi-collinearity were performed, to minimize the number of inputs that need to be measured. The three hybrid models perform similarly in the test experiments in dataset A, with an NRMSE of 0.07. In dataset B, predictions had slightly higher NRMSE (less than 0.02), with RFs performing the best. Authors also investigated performance when training the hybrid models in dataset B; however, they observed that the cumulative amount of glucose predictions did not improve, contrary to viable cell density and titer.

Neural ODEs are used as expansion for non-accounted dynamics in the structure of parametric models [214]. The authors carried out experimental fed-batch cultivations using an engineered β -carotene producing *S. cerevisiae* strain. Over 50 parametric model structures were tested, identifying the best one. After identifying the best parametric structure, the Neural ODEs were implemented as an additive term to the ODEs describing biomass growth and ethanol production. The hybrid model exhibited 3.6-fold and 1.7-fold better performance in the training and testing datasets, when compared to the best parametric model structure. Despite longer training and calibration times than the parametric model, the hybrid model executed faster.

There is a growing interest in applying hybrid modeling for commercial and industrial uses, with AspenTech filing for a patent [252, 253], the foundation of companies specialized in the area like DataHow AG and Novasign, and a recent interest on behalf of the pharmaceutical industry [59, 99, 101, 102].

4.6 | Synthesis of Key Findings

Hybrid models in bioprocesses predominantly employ ANNs as the data-driven component, accounting for 58% of publications. ANNs have been widely applied across various bioprocesses, from cell cultivation to pharmaceutical production. Other ML techniques, such as SVMs, have also been utilized, particularly in cell cultivation and WTT. In lesser proportion, parametric techniques such as polynomials and sigmoidal expressions have been applied successfully to metabolic fluxes analysis [136], and lutein production [139]. From an application perspective, various categories of data-driven techniques have been successfully employed in bioprocessing. However, there is no consensus on the most suitable method for each process type, despite a clear preference for ANNs.

Over 63% of work targets metabolite production—chiefly fermentation and protein synthesis—while ~40% address pharmaceutical operations (penicillin, therapeutic proteins, antibodies, CHO cells) [101, 102]. As such, the enhanced prediction capabilities of hybrid models combined with specific approaches to experimental design [59, 98, 99] have led to innovation in the field. WTT comprises ~10% of cases but leads in industrial uptake, underscoring hybrid models’ aptitude for revealing hidden dynamics under sparse monitoring.

In terms of hybrid model architecture, feature learning hybrids dominate despite offering no statistically significant performance advantage over delta or serial patterns. Their popularity likely stems from built-in interpretability, conservation law enforcement, and modular training, which suit the multi-step, partially observed nature of bioprocesses [20].

Ethanol fermentations of *S. cerevisiae* remain the most intensely studied system—driven by abundant data—while recent work has diversified to include *E. coli*, *P. pastoris*, CHO cells (collectively 30%), and a broad “other” category (32%). Studies equally showcase hybrid models across batch (44.6%), fed batch (33.7%) and continuous (20.7%) modes of operation and at lab (53%), pilot (25%) and industrial (21.7%) scales, underscoring their adaptability [119].

Software choice and model evaluation are areas of persistent inconsistency. MATLAB remains the predominant simulation environment, although Python is gaining traction. Model performance assessment still is variant: about a quarter of studies still rely on r^2 —despite its failure to penalize over parameterization and insensitivity to nonlinear dynamics [94, 254]. On the other hand, 45% of contributions use MSE derivatives to calculate their performance indexes, despite it being highly sensitive to outliers [255]. This last selection might be since it is a default performance index in ML toolboxes. To foster more robust and interpretable assessments, we foresee that these performance indexes could be complemented with information-theoretic and distribution-based measures (e.g., entropy coefficients, Kolmogorov–Smirnov statistics, Log-Cosh and quantile losses) and adopting principled model-comparison criteria such as Watanabe Akaike Information Criterion (WAIC) or Leave-One-Out Cross-Validation Information Criterion (LOO-CVI) [256–259].

Despite three decades of development, hybrid bioprocess modeling still lacks standardized protocols for selecting design patterns, balancing mechanistic and data-driven components, quantifying uncertainty and defining data preprocessing workflows. The field's reliance on trial and error and experience—evidenced by 374 individual model implementations versus 155 distinct case studies—underscores this gap. Some efforts have been made recently to highlight practical and technical details of implementing hybrid models, such as reviews that present brief guides for developing them [25, 260], but more work is needed. Due to having both a parametric/mechanistic component and a data-driven/nonparametric component, assessing how much uncertainty comes from each remains an unexplored avenue. The field could benefit from recent developments in the ML field, where authors suggest a Monte Carlo approach to generate testing datasets and better capture uncertainty [261].

The absence of standardized guidelines in hybrid bioprocess modeling arises precisely from its defining strength: structural flexibility. Hybrid frameworks can accommodate a wide range of design choices—yet still outperform purely mechanistic models—while demanding far less data than fully nonparametric approaches and preserving a degree of interpretability that renders them practical for industrial deployment. However, this very hybrid character also complicates evaluation: conventional metrics and workflows developed for parametric or nonparametric models often fail to capture the unique tradeoffs in uncertainty quantification and mechanistic parameter estimation inherent to hybrid systems.

Numerical workflows in hybrid bioprocess modeling are similarly under specified. Critical steps—data pretreatment methods (both numerical and knowledge based), augmentation strategies, and the rationale for selecting inputs to the data driven module—are frequently omitted, leaving reproducibility and comparability in doubt. Moreover, cross-validation protocols and data partitioning schemes are rarely described, despite their importance for robust training and unbiased assessment of models that rely heavily on high-quality data to capture complex, industrially relevant dynamics. To address these gaps, the community should adopt rigorous information-theoretic tools

to quantify intervariable relationships and guide model structure, and standardize transparent cross-validation procedures to ensure reliable, reproducible hybrid model development [262].

Cross validation offers clear advantages—correcting selection bias, quantifying estimation uncertainty (particularly for non-parametric components) and guiding model complexity to prevent overfitting [263]. Yet, despite its value, consensus on hyperparameter tuning remains elusive. While many practitioners adhere to the conventional three-way split (training, validation, test) coupled with grid, random or Bayesian searches, limited data often forces a two-set approach (training and validation only). Without systematic cross validation, hybrid models inherit latent selection bias, further obscuring true predictive uncertainty. Although a handful of specialized diagnostic tools exist for assessing hybrid performance, the community still lacks standardized protocols for data partitioning and hyperparameter optimization—an important frontier for improving both transparency and reliability in hybrid bioprocess modeling.

The challenges outlined above often originate in insufficient data quantity and quality, underscoring the necessity of integrating hybrid-model data requirements into the experimental design phase. In this context, Intensified Design of Experiments (iDoE) offers a compelling solution: by strategically varying inputs and conditions, iDoE minimizes the total number of trials while maximizing the informational content of each run, thereby furnishing richer datasets that better support the calibration and validation of hybrid bioprocess models [59, 98, 99, 117]. Of course, this strategy is also limited by monitoring capabilities, since online data acquisition of primary variables at high frequency still is a challenge. Developments in process analytical technologies (PAT) definitely will catalyze the advanced modeling strategies as hybrid models.

4.7 | Hybrid Modeling Challenges

Certain challenges for implementing hybrid models have been mentioned in the literature [26, 55, 264]:

1. The mechanistic part of the hybrid model constrains the space of solutions obtainable, which helps avoid pitfalls such as negative concentrations. These constraints impose an inductive bias into the responses, which might negatively affect the hybrid model's performance if the mechanistic part is not suitable for describing the process.
2. There are standard techniques for identifying parameters in statistical or purely parametric models. For hybrid models, only a handful of parameter identification methods have been reported [57, 95, 104, 265]. The major hurdle is the time needed for parameter identification, due to the extensive calculations that should be performed.
3. In addressing the initial challenge, the integration of hybrid models requires a thorough examination of the foundational assumptions inherent in the mechanistic knowledge to be integrated. This task can be approached via two distinct methodologies:

- a. The incremental approach, where the most basic assumptions are made, and more complex interactions are added until the model performance decreases.
- b. The removal approach, where the most complex fundamental structure is evaluated first, and then features are removed iteratively to determine what part is hindering the model's performance.

Both approaches are time consuming, so model development will require some experience or an analysis of a risk of mechanistic knowledge being incorrect.

4. Validating and verifying a hybrid model is another challenge, because there needs to be a balance of complexity, accuracy, interpretability and scalability which is hard to optimize. Moreover, in hybrid models, nonparametric and parametric components can introduce distinct sources of uncertainty and error that must be carefully managed [26].
5. The level of “grayness” in hybrid models—the balance between mechanistic knowledge and data-driven components—can significantly impact performance. As illustrated by Rogers et al. [183], incorporating too much or too little mechanistic knowledge can lead to bias or overfitting, respectively, thus hindering model performance. This underscores the need for meticulous calibration and validation of hybrid models to ensure optimal balance and performance. Regardless of advancements in key areas, still there is not a unified framework for determining the valid region for extrapolation of hybrid models.

Besides the challenges addressed before, we propose the following ones that emerge from the literature review:

1. There are no consumer-ready applications built on open-source software for building hybrid models easily, which hinders its adoption by industry, due to the need for specialized personnel to build, operate and maintain the models. Although some custom toolboxes for MATLAB like HYBMOD [125, 220, 222, 233] exist, source code is poorly documented, and a licensing agreement should be signed.
2. As with many pure data-driven techniques, there is not a consolidated framework that allows to select the proper nonparametric technique a priori, meaning that a costly process of trial and error is necessary to build hybrid models.
3. Like the previous challenge, there is a lack of a general and unified framework for deciding which type of hybrid model structure/design pattern is best suited for an application, which also results in a resource-intensive search.
4. As with many nonparametric techniques, how data should be split between training, testing and validations has no consensus. Authors commonly only use training and validation splits due to constraints on the availability of high-quality data, while few report using datasets for the three purposes.
5. There is no consensus on the training hierarchy of the parametric and nonparametric components of hybrid models. When using the “delta model” design pattern, the training of each subcomponent is done separately. However, when

using the other design patterns, there are no clear guidelines on how to address the identification of the parametric components and the nonparametric components of hybrid models. Some authors choose the best pure parametric model structure available, start hybridizing and then calibrating the nonparametric structure over it [214]. Other authors do the calibration of both components at the same time [266, 267]. Whether one approach yields better results than the other needs to be further investigated.

4.8 | Hybrid Modeling Perspectives

Looking forward, several promising avenues for future research and development in hybrid semi-parametric modeling are clear. One key area is the continued refinement of model integration techniques. Advancements and applications of deep learning [125, 220, 222] and the development of sophisticated neural network architectures offer opportunities to enhance the nonparametric components of hybrid models. The application of Bayesian inference, could improve the accuracy and confidence of predictions by incorporating uncertainty quantification into the modeling process [248].

Additionally, expanding the application of hybrid models to new bioprocesses and products will be crucial. As the bioprocessing industry continues to innovate and develop new products, hybrid models will need to adapt and evolve to meet these emerging challenges. The growing interest from commercial and industrial sectors, including companies like AspenTech and DataHow AG, as well as the pharmaceutical industry, further underscores the relevance and applicability of hybrid models in real-world settings. The development of standardized frameworks and best practices for hybrid model implementation could facilitate broader adoption and integration into industrial processes.

Authors should remain open-minded and explore solutions and implementations beyond hybrid modeling in bioprocesses or applied mathematics or computer science; for instance, disciplines like agriculture and meteorology have yielded promising hybrid modeling strategies [268–270]. Although implementations are dependent on data quality and specific to each field, general insights and strategies can be adapted.

Hybrid semi-parametric models represent a powerful tool for advancing bioprocess modeling and optimization. Integrating mechanistic and data-driven approaches offers a pathway to more accurate, reliable, and scalable models that can meet the complex demands of modern bioprocessing. Continued research and innovation in this field will undoubtedly unlock new opportunities and enhance the efficiency and effectiveness of bioprocesses across various industries.

5 | Concluding Remarks

This work presents a comprehensive framework for hybrid modeling in bioprocesses, drawing insights from 270 references. It covers hybrid models' characteristics, architecture, and nonparametric structures. Hybrid models are primarily

applied to metabolite production, especially ethanol and penicillin fermentations—followed by cell cultivation and WTT. In protein production, antibody manufacturing is significant. ANNs are the dominant nonparametric method, though other ML techniques such as random processes and genetic algorithms are also used. Most models are trained on experimental data from laboratory-scale studies, with some full-scale industrial applications. Batch and fed-batch systems are the most common operation modes, while continuous processes are less explored. Frequently used microorganisms include *Saccharomyces cerevisiae*, *E. coli*, and CHO cells. MATLAB is the preferred software, though Python is gaining popularity due to its robust ML toolboxes. Finally, while MSE and related metrics are common, no consensus exists on how to best evaluate hybrid model performance.

PSE provides the critical link between these hybrid-model capabilities and global sustainability imperatives. By embedding life-cycle assessment, techno-economic analysis and multi-objective optimization within hybrid modeling workflows, PSE enables quantitative trade-off studies that reconcile yield, energy consumption, emissions and cost from early design through scale-up. Digital-twin simulations and real-time soft sensors further ensure adaptive control and resilience under variable feedstocks. Such integrated PSE-driven approaches directly support UN SDG 7 (clean energy), SDG 9 (industry and innovation), SDG 12 (responsible production) and SDG 13 (climate action) by valorizing biomass waste, minimizing environmental impact and fostering low-carbon biomanufacturing. Standardizing PSE-embedded sustainability metrics within hybrid models will be pivotal for accelerating tangible SDG progress and closing the financing and implementation gaps that currently jeopardize the 2030 agenda.

Author Contributions

Juan Federico Herrera-Ruiz: conceptualization, data curation, methodology, software, investigation, formal analysis, visualization, writing – original draft, writing – review and editing. **Javier Fontalvo:** supervision, writing – review and editing. **Oscar Andrés Prado-Rubio:** supervision, project administration, funding acquisition, writing – original draft, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that supports the findings of this study are available in the [Supporting Information](#) of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** eng270502-sup-0001-supinfo.xlsx.